

## Elisa Bortolas – Curriculum Vitae

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### Professional Experience

**Maternity leave: 2023-2024, 8 full months**, during which I was the primary caregiver of my child and did not work.

- 01/10/2026, ongoing: permanent researcher, Istituto Nazionale di Astrofisica, Osservatorio Astronomico di Padova (Italy)
- 01/10/2024–30/09/2026: fixed term research position, Istituto Nazionale di Astrofisica, Osservatorio Astronomico di Padova (Italy)
- 01/11/2023–30/09/2024: **Marie Skłodowska-Curie fellow**, University of Milano-Bicocca
- 01/11/2020 – 31/10/2023: Postdoctoral research fellow funded through the European Research Council (ERC), University of Milano-Bicocca, Italy
- 01/04/2019 – 31/10/2020: Postdoctoral research fellow, Institute for Computational Science, University of Zürich
- 01/11/2018 – 31/03/2019: Research Assistant, Institute for Computational Science, University of Zürich

### Education

- 01 Oct 2015 – 30 Sept 2018: **Ph.D. in Astronomy** awarded by the University of Padova (Italy) on March 5<sup>th</sup>, 2019. Thesis: *Dynamics of Single and Binary Black Holes in Galactic Nuclei*; Supervisor: Prof M. Mapelli; co-Supervisor: Dr A. Gualandris.
- 01 Nov 2013 – 21 Sept 2015: **Master's degree in Astrophysics and Space Physics** awarded by the University of Milano-Bicocca (Italy) on September 21<sup>st</sup>, 2015; mark 110/110 **cum laude**. Thesis: *Dynamics of Massive Black Hole Binaries in Galactic Merger Remnants*; Supervisor: Prof M. Dotti; external supervisor: Dr A. Gualandris.
- 01 Sept 2010 – 21 Oct 2013: **Bachelor's degree in Physics** awarded by the University of Milano-Bicocca (Italy) on October 21<sup>st</sup>, 2013; mark 110/110 **cum laude**. Thesis: *Analysis of a sample of mega-maser hosting galaxies*. Supervisor: Prof M. Dotti; co-supervisor: Prof G. Gavazzi.

### Extended research visits

*A total of about 18 months of research visits in leading international research institutes*

- 2023, February-March: Visit to the **Donostia International Physics Centre**, San Sebastian (Spain) to develop a project in collaboration with Dr S. Bonoli and M. Polkas (**1 month**)
- 2022, December: Visit to the **Laboratoire des 2 Infinis** in Toulouse (France) to develop a project in collaboration with Dr M. Toscani and Dr N. Tamanini (**2 weeks**)
- 2022, March-April: **Kavli Institute for Theoretical Physics** program *Bridging the Gap: Accretion and Orbital Evolution in Stellar and Black Hole Binaries*, University of California – Santa Barbara, US (**3 weeks**).
- 2017: Visiting PhD Student at the **University of Surrey**, UK (**10 months**).
- 2015: Erasmus+ Traineeship Programme (Research Internship) at the **University of Surrey**, UK (**6 months**).

## PRIZES, AWARDS, GRANTS

### Fellowships and research funds

- 2026: **Curiosity Science Grant** funded by INAF (call AF2025); awarded as Principal Investigator (PI) for a 2-year high-risk, seed-funding project to test the feasibility of innovative, larger scientific projects.
- 2023: **Marie Skłodowska-Curie Postdoctoral Fellowship – Global Fellowship**
- 2016: **Ing. Gini Fellowship** funded by the Italian Aldo Gini Foundation; the fellowship supports a period of research and education outside Italy for Italian STEM graduates

2015: Awarded funding for an **Erasmus+ Traineeship** program; this EU funding scheme promotes training and education through the mobility of students among European institutions

### Honors and awards

- 2025: Awarded the **Italian National Scientific Qualification** (Abilitazione Scientifica Nazionale – ASN), a prerequisite for appointment as Associate Professor in Italian universities
- 2023: **Honourable mention, ‘Premio Giovani Ricercatori’** (Young Researchers prize – Astrophysics and Space category) acknowledging the excellence of a scientific publication led by an early-career researcher (Bortolas et al., *The Competing Effect of Gas and Stars in the Evolution of Massive Black Hole Binaries*, 2021, ApJL 918, L15).
- 2021: **‘Giovani Talenti’ prize** acknowledging young researchers with an excellent publication record; 3<sup>rd</sup> place; awarded by the University of Milano-Bicocca and the *Accademia dei Lincei* (the most prestigious scientific academy in Italy, and the oldest in the world).
- 2019: **Livio Gratton prize for the best Italian PhD thesis in Astrophysics** in 2017 – 2018
- 2018: **Honourable mention, GWIC-Braccini PhD Thesis Prize**; international yearly prize for the best PhD thesis in the field of gravitational waves ([gwic.ligo.org/thesis-prize.html](http://gwic.ligo.org/thesis-prize.html)).
- 2017: **Stefano Magini Prize for the best Italian Master thesis in Astrophysics** in 2015 – 2016;
- 2017: **Best poster prize** awarded at the 11th TRR33 Winter School in Cosmology, Theory for Observers & Observations for Theorists (Passo del Tonale, Italy, December 2017; poster title: *Supernova Kicks and Dynamics of Compact Remnants in the Galactic Centre*)

### Funding record

In short, obtained a total of *more than 325,000 EUR of funding from different sources*.

- 2026: **Curiosity Science Grant**, AF2025 **15,000 EUR**
- 2023: **Marie Skłodowska-Curie Postdoctoral Fellowship – Global Fellowship**, **288,859 EUR**
- 2023: **AHEAD AO4 Visitor Program**, awarded **3,000 EUR**
- 2022: **EuCAPT** exchange programme, awarded **1,800 CHF** (~1,820 EUR)
- 2021: **‘Giovani Talenti’ (Young Talents) prize** at the University of Milano-Bicocca **1,000 EUR**
- 2019: **Gratton Prize** for the best Italian PhD thesis in Astrophysics 2017 – 2018, **5,000 EUR**
- 2017: **Magini Prize** for the best Italian master thesis in Astrophysics 2015 – 2016, **1,250 EUR**
- 2016: **Ing. Gini Fellowship** to support a period of study or research abroad, **6,400 EUR**
- 2015: **Erasmus+ Traineeship** to support a research period in a foreign EU institution, **3,000 EUR**
- 2015–now: Granted about **3,400 EUR** (cumulative) for participating in international scientific meetings, schools and research visits, *plus granted cumulative board and lodging for a total of 36 days/nights*.

## MENTORING AND TEACHING EXPERIENCE

### Teaching experience

- 2025-2026: **Main lecturer** for the PhD-level course *Computational Astrophysics*; the course is held for the PhD programme in *Astronomy* in the Physics and Astronomy Dept. of the University of Padova.
- 2025: **Invited Lecturer for the doctoral school** *LISA School for early-career scientists*, Les Houches School of Physics (France), October 6–17, 2025.
- 2022-2023: **Teaching assistant** for the master-level course *Galaxies and Dynamics* (blended learning); the course is held for the master degree in *Astrophysics and Space Physics* in the Physics Dept. of the University of Milano-Bicocca.
- 2023: **Invited Lecturer for the doctoral school** *Nordic Winter School on Gravitational Astrophysics*, Skeikampen (Norway), January 29 – February 3, 2023. Lectures topic: *Formation and evolution of super-massive black hole binaries*
- 2021-2022: **Main Lecturer** for the master and PhD-level course *Scripting and Programming Laboratory for Data Analysis* held for the Department of Science and High Technology of the University of Insubria, Como (Italy). *I taught a total of 33 hours. I prepared most of the course structure and lectures content,*

*including the exercises; I developed the structure of the final exam; I evaluated all final exams and ranked the students' performance.*

2018-2019 AND 2019-2020: **Guest Lecturer** and **Teaching Assistant** for the master-level *Computational Astrophysics Lab* course, U. of Zürich; *I entirely developed and taught the part of the course related to the Direct Summation Technique; I developed a substantial part of the material used for the students' final project; I participated in the exams and in the marking process.*

2018-2019 AND 2019-2020: **Teaching Assistant**, Course *The Universe: Contents, Origin, Evolution and Future*, U. of Zürich; *I prepared the exercises and I taught the exercise lessons; I assisted the students developing the final project following the guidelines I outlined; I took part in all exams and extensively participated in the marking process.*

2018: Granted 24 European University Credits (CFU) for teaching in secondary school, U. of Padova (Italy).

## Students co-supervision and co-mentoring

In short, I **supervised, co-supervised or mentored**: *6 PhD students, 11 master students, 2 bachelor students, 2 other.* **Note:** Since 2024, my affiliation with a research institute offers fewer opportunities for student recruitment compared to a university setting.

*PhD students:* Lorenz Zwick (U. of Zürich, 2019 – 2020; main supervisor: L. Mayer), Rafael Souza-Lima (U. of Zürich, 2019 – 2020; main supervisor: L. Mayer), Verónica Vázquez Aceves (Peking U., 2020–2021; main supervisor: X. Chen), Ludovica Varisco (U. of Milano-Bicocca, 2020–2023; main supervisor: M. Dotti), Luca Broggi (U. of Milano-Bicocca, 2020–2023; main supervisor: A. Sesana), Federica Fastidio (U. of Milano-Bicocca & U. of Surrey – UK, 2022–2025; main supervisors: A. Sesana & A. Gualandris)

*Master students:* Boris Pestoni (U. of Zürich, 2019; main supervisor: L. Mayer), Lorenz Zwick (U. of Zürich, 2019; main supervisor: L. Mayer), Nikolas Wittek (U. of Zürich, 2019; main supervisor: L. Mayer), Jakob Stegmann (U. of Zürich, 2019; main supervisor: L. Mayer), Ludovica Varisco (U. of Milano-Bicocca, 2020; main supervisor: M. Dotti), Giada Montini (U. of Milano-Bicocca, 2021; main supervisor: M. Colpi), Martin Pijnenburg (U. of Zürich, 2021; main supervisor: L. Mayer), Matteo Bisiani (U. of Milano-Bicocca, 2021–2022; main supervisor: M. Dotti), Vittorio Visconti (U. of Milano-Bicocca, 2024; main supervisor: A. Sesana), Giulio Gozzi (U. of Milano-Bicocca, ongoing; main supervisor: A. Sesana), Martina Gallieni (U. of Milano-Bicocca, ongoing; main supervisor: M. Colpi)

*Bachelor students:* Andrea Incatasciato (U. of Milano-Bicocca, 2015; main supervisor: M. Dotti), Pietro Zeduri (U. of Milano-Bicocca, 2023; main supervisor: A. Sesana)

*Introduction to research experience:* Tobia Vecchiatto Montorio (U. of Padova, 2025), Gaia Fittipaldi (U. of Padova, 2025)

## SEMINARS AND TALKS

### Invited seminars and talks at international research institutes

In short: *16 invited presentations*

16. Invited seminar at the Institute for Computational Science, **University of Zürich** (CH), September 21, 2023

15. Invited seminar at the **Donostia International Physics Centre**, San Sebastian (Spain), March 8, 2023

14. Invited seminar at the **University of Surrey** (UK), February 28, 2023

13. Invited seminar at the *Department of Applied Mathematics and Theoretical Physics* of the **University of Cambridge** (UK), February 27, 2023

12. Invited seminar at the **Institut de Recherche en Astrophysique & Planétologie** in Toulouse (France), December 16, 2022

11. Invited seminar at the **Laboratoire des 2 Infinis** in Toulouse (France), December 8, 2022

10. Invited talk at **Columbia University**, New York (US), November 3, 2022

9. Invited colloquium at **Harvard University** (US), *Black Hole Initiative Colloquium series*, October 31<sup>st</sup>, 2022

8. Invited seminar at **Clemson University** (US), *Summer Seminar Series on Gas Dynamics and Gravity*, July 8<sup>th</sup>, 2022

7. Invited seminar at the **Kavli Institute for Theoretical Physics** in Santa Barbara (US), April 7<sup>th</sup>, 2022

6. Invited talk for the **NANOGrav collaboration** astrophysics working group, November 4<sup>th</sup>, 2021
5. Invited seminar for the **Laura Bassi Seminar Series** organized by the INAF (Italian National Institute for Astrophysics), February 11<sup>th</sup>, 2021
4. Invited seminar at the **University of Zürich** (Switzerland), January 25<sup>th</sup>, 2019
3. Invited seminar at the **University of Innsbruck** (Austria), March 12<sup>th</sup>, 2018
2. Invited talk at the **University of Zürich** (Switzerland), November 27<sup>th</sup>, 2017
1. Invited talk at the **INAF – Osservatorio Astronomico di Arcetri** (Italy), May 30<sup>th</sup>, 2017; invited talk for the award of the *Stefano Magini* Master thesis prize

### Conferences and Workshops: summary and selected contributions

In short: *20 invited talks, 17 contributed talks, 2 poster presentations + 1 at a PhD school, 4 discussion moderations, chair of 3 talk sessions; invited talks in red.*

33. SEXTEN WORKSHOP, *Climbing Cosmic Peaks*, March 2–6, 2026 at the Sexten Centre for Astrophysics; **invited talk** *MORFEO@ELT in the context of future massive black hole (binaries) observations*
32. OAPD DAY, June 20, 2025 at the Asiago Astrophysical Observatory; **invited talk** *The path to coalescence of supermassive black hole binaries across time and space*
31. MORFEO SCIENCE TEAM MEETING WORKSHOP, May 27–29, 2025 in Bologna; **talk** *Galactic core-scouring by supermassive binaries with MORFEO*
30. SEXTEN WORKSHOP, *Frontiers of Astrophysical Black Holes*, March 17–21, 2025 at the Sexten Centre for Astrophysics; **invited talk** *Core-scouring in galaxies and black hole binary merger rates*
29. IFPU FOCUS WEEK PROGRAM ON “THE QUEST FOR DUAL AND BINARY SUPERMASSIVE BLACK HOLES IN THE MULTI-MESSENGER ERA”, January 13–17, 2025 at the Institute for Fundamental Physics of the Universe (IFPU) in Trieste (Italy); **invited review talk** *Binary supermassive black holes on the theoretical side*
28. BLACK HOLES ON BROADWAY, December 4–7, 2023 at the Center for Computational Astrophysics (CCA) of the Flatiron Institute in New York; **invited review talk** *The modelling of massive black hole mergers and dynamics*
27. YAGN23 MEETING, October 18–20, 2023 at the Istituto di Astrofisica Spaziale e Fisica Cosmica in Palermo, Italy; **invited talk** *On the importance of characterizing partial stellar disruption event rates in the era of LSST*
26. 4TH GRAVI-GAMMA-NU WORKSHOP, October 4–6, 2023 at the Gran Sasso Science Institute (GSSI) in L’Aquila (Italy); **invited review talk** *Tidal disruption event rates under the magnifying glass*
25. EUROPEAN ASTRONOMICAL SOCIETY MEETING, July 10–14, 2023 in Krakow, Poland; **2 invited review talks**, one for the *Symposium 12: Breaking down the AGN frontiers – Light and sound from early black holes*, the other for the *Special Session 17: The Last Kiloparsecs: From Dual AGN to Massive Black Hole Binary Mergers*; **1 talk** for the *Special Session 35: Supermassive Black Hole Transients and Their Host Galaxies*
24. LUNAR GRAVITATIONAL WAVE ANTENNA WHITE PAPER KICK-OFF MEETING, February 9–10, 2023 at the Gran Sasso Science Institute, L’Aquila (Italy); **invited review talk** *Tidal disruption events and the LGWA*
23. SEXTEN WORKSHOP, *Getting ready to descend the slippery slope of multimessenger cosmological black holes data*, January 23–27, 2023 at the Sexten Centre for Astrophysics; **invited talk** *Partial tidal disruption events*
22. YAGN22 MEETING, October 17–19, 2022, at the Donostia International Physics Center (Spain); **talk** *Tidal disruption event rates and the importance of a complete stellar mass function*
21. 14TH INTERNATIONAL LISA SYMPOSIUM, July 25–29, 2022, online meeting; **talk** *Gas, stars and gravitational waves: on the main driver of massive black hole binaries path to coalescence*
20. LISA ASTROPHYSICS WORKING GROUP MEETING, June 22 – 24, 2022 in Birmingham, UK:
  - **invited review talk** *Astrophysics with LISA: massive black holes*
  - **talk** *Gas, stars and gravitational waves: on the main driver of massive black hole binaries path to coalescence*

19. Meeting INTERMEDIATE-MASS BLACK HOLES: NEW SCIENCE FROM STELLAR EVOLUTION TO COSMOLOGY, April 30 – May 3, 2022, in San Juan, Puerto Rico; **talk** *Tidal disruption event rates and the importance of a complete stellar mass function*
18. GW NEXT MEETING, January 12–14, 2022, online meeting; **invited review talk** *Dynamical evolution of supermassive black hole binaries across time and space*
17. EPTA WINTER MEETING 2021, December 6–8, 2021, online meeting; **talk** *Stars vs gas: on the main driver of massive black hole binaries evolution prior to gravitational waves*
16. YAGN21 MEETING, September 1–3, 2021, at the Niels Bohr Institute – University of Copenhagen (Denmark); **invited talk** *Stars vs gas: on the main driver of massive black hole binaries evolution prior to gravitational waves*
15. TEONGRAV online workshop, February 18, 2021; **talk** *Dynamical evolution of supermassive black hole pairs in realistic galaxy models*
14. SUPERMASSIVE BLACK HOLES conference, December 7–11, 2020, online meeting; **talk** *Global torques and stochasticity as the drivers of massive black hole pairing at  $z \sim 6$* ; **discussion moderator** of the session *Black hole binaries from LIGO to LISA*
13. YAGN20 MEETING, October 28–30, 2020, online meeting; **invited talk** *Curse or blessing? The role of bars in the large-scale pairing of massive black holes*
12. 13TH INTERNATIONAL LISA SYMPOSIUM, September 1–3, 2020, online meeting; **talk** *The role of bars and spirals on the pairing of massive black holes at  $z \sim 6$*
11. LISA ASTROPHYSICS WORKING GROUP WORKSHOP, March 9–11, 2020 in Nijmegen (Netherlands), **talk** *The stochastic pairing of massive black holes in the young Universe*
10. SEXTEN WORKSHOP, *Getting ready to descend the slippery slope of multimessenger cosmological black holes data*, February 10–14, 2020 at the Sexten Centre for Astrophysics (Italy); **invited talk** *The stochastic pairing of massive black hole binaries in the young Universe*
9. YAGN19 MEETING, September 23–25, 2019 in Tenerife (Spain); **invited summary talk of the theoretical presentations** and **invited talk** *Evolution of supermassive black hole pairs at  $z \sim 6$*
8. TAL ALEXANDER MEETING 2019, August 25 – September 5, 2019 in Alájar (Spain); **invited blackboard presentation** on the dynamics of extreme mass ratio inspirals
7. 1<sup>st</sup> LISA ASTROPHYSICS WORKING GROUP WORKSHOP, December 12–14, 2018 in Paris (France); **talk** *Can supernova kicks trigger EMRIs in the Galactic Centre?*
6. YAGN18 MEETING, October 29–31, 2018 in Budapest (Hungary); **invited talk** *Can supernova kicks trigger EMRIs in the Galactic Centre?*
5. MODEST-18 MEETING, June 25–29, 2018 in Santorini (Greece); **talk** *EMRIs Triggered by Supernova Kicks in the Galactic Center*
4. GRAVITATIONAL-WAVE SCIENCE & TECHNOLOGY SYMPOSIUM 2018, March 1 – 2, 2018 in Padova (Italy); **talk** *Star Cluster Disruption by a Super-massive Black Hole Binary*
3. MODEST-17 MEETING, September 18 – 22, 2017 in Prague (Czech Republic); **talk** *Star Cluster Disruption by a Supermassive Black Hole Binary*
2. LEMAITRE WORKSHOP on Black Holes, Gravitational Waves and Spacetime Singularities, May 9 – 12, 2017, Vatican Observatory, Albano Laziale (Italy); **talk** *Massive Black Hole Binaries and their Hosts: a common journey towards Gravitational Waves.*
1. MODEST-15s MEETING, December 7–11, 2015 in Kobe (Japan); **talk** *Dynamics of Massive Black Hole Binaries in Galactic Merger Remnants.*

## PI-SHIPS AND COI-SHIPS

### Accepted observational and computational proposals

- 2025: **PI** of the proposal *Stellar tidal disruption events onto intermediate mass black holes* at CINECA, **80k GPU+CPU hours** awarded for  $N$ -body simulations on the Leonardo Booster cluster.
- 2023: **Co-I**, XMM-Newton AO23 proposal, *Testing star-disk EMRI models with the well-behaved QPE source eRO-QPE2*, 146 ks awarded by the Observing Time Allocation Committee, PI: Riccardo Arcodia.

- 2022: **Co-I**, Rapid Eye Mount (REM) DDT proposal, *Rapid evolution of a binary massive black hole system*, PI: Massimo Dotti, ROS2 (10h), REMIR (10h).
- 2021: **Co-I (the PI, L. Varisco, is a student I mentored and guided in the application)** of the proposal *Evolution of a Massive Black Hole Binary in Rotating Stellar Systems* at CINECA, **32k CPU hours** awarded for  $N$ -body simulations on the MARCONI100 cluster.
- 2020: **Co-I** of the proposal *The PHOEBOS: formation and evolution of the first massive black holes* at CSCS, about **15M CPU hours** awarded for SPH simulations on the hybrid CPU+GPU partition of the PizDaint cluster.
- 2016: **PI** of the proposal *Evolution of Massive Black Hole Binaries in their host systems* at CINECA, about **100k CPU hours** awarded for  $N$ -body simulations on the GALILEO IBM cluster.
- 2015: **PI** of the proposal *Evolution of Massive Black Hole Binaries* at CINECA, about **100k CPU hours** awarded for  $N$ -body simulations on the GALILEO IBM cluster.

### Accepted proposals for research visits and scientific meetings

- 2023: **Co-I** for the proposal of a Symposium for the **European Astronomical Society Annual Meeting 2024**; proposal title: “*Lighthouses on the Universe: astrophysics and cosmology with Pulsar Timing Arrays*”.
- 2023: **PI (no Co-Is)** for the proposal for a research visit funded by the **AHEAD Announcement of Opportunity 4 – Visitor program**; the program funded a visit at the *Donostia International Physics Centre* in San Sebastian (Spain)
- 2022: **PI (no Co-Is)** for the proposal for a research visit funded by the **EuCAPT exchange programme**; the program funded a visit at the *Laboratoire des 2 Infinis* in Toulouse, France
- 2020: **Co-PI** for the proposal of a Special Session for the **European Astronomical Society Annual Meeting 2021**; proposal title: “*The route to coalescence of supermassive black hole binaries: a comprehensive view*”.

## OTHER ITEMS OF INTEREST

### Organization of scientific conferences and meetings

- 2025: **Organizing Committee member** for the meeting “*Loss Cones in Como*” in Como, Italy, September 15–17, 2025
- 2024: **SOC member** for the organization of the Symposium “*Lighthouses on the Universe: astrophysics and cosmology with Pulsar Timing Arrays*” at the European Astronomical Society Meeting 2024 (Padova, Italy)
- 2024: **Organizing Committee member** for the meeting “*Loss Cones in Como*” in Como, Italy, February 6–8, 2024
- 2023: **LOC member** for the organization of the “*LISA Astrophysics Working Group 2023*” in Milan, Italy, September 13–15, 2023
- 2023: **LOC member** for the organization of the “*EPTA 2023 spring meeting*” in Milan, Italy, March 13–17, 2023
- 2022: **SOC and LOC Chair** (together with M. Dotti) for the organization of the meeting “*Tidal disruption event and extreme mass ratio inspiral dynamics*” in Como, Italy, September 5–7, 2022.
- 2021: **SOC Chair** (together with M. Bonetti) **for the organization of the Special Session** “*The route to coalescence of supermassive black hole binaries: a comprehensive view*” at the European Astronomical Society Meeting 2021 (online)

### Institutional responsibilities and Service

- 2026: **Selection Board for a 12-month Research Fellowship** on “*Design, Implementation, and Testing of Real-Time Computers for Adaptive Optics in Astrophysics*” at INAF-Osservatorio Astronomico di Padova.
- (year not to be disclosed) Member of the **panel review for the NASA Astrophysics Theory Program (ATP)** solicitation
- (year not to be disclosed) Member of the **panel review for the NASA ROSES LISA Preparatory Science** solicitation

- 2024: **Editor of the book** *Black Holes in the Era of Gravitational-Wave Astronomy* (together with M. Arca-Sedda and M. Spera), published by Elsevier in May 2024
- 2023–2024: Mentor for the LECS-organized *LISA mentorship program*
- 2023: Main organizer (of 5) of the survey *What is the status of equality, diversity and inclusion in the Astrophysics Group at University of Milano-Bicocca?* and organizer of a series of events for improving the working climate in the Astrophysics group at Milano-Bicocca.
- 2023: **Coordinator for the drafting** of the part focusing on *Populations and formation channels of GW sources* for the White Paper of the **Lunar Gravitational Wave Antenna**
- 2021– 2022: Organizer of the weekly group meetings of the B Massive research group at the U. of Milano-Bicocca; main manager of the group website <https://sites.google.com/unimib.it/bmassive/>
- 2018 – now: **Reviewer for** *Nature Astronomy* (NA), *the Astrophysical Journal* (ApJ), *Monthly Notices of the Royal Astronomical Society* (MNRAS), *Astronomy & Astrophysics* (A&A), *Publications of the Astronomical Society of Japan* (PASJ) on a regular basis.
- 2020–2021: **Coordinator for the drafting** of Chapter 2 of the **LISA Astrophysics Working Group White Paper**
- 2019 – 2020: Organizer of the Seminars for the Institute of Computational Science, U. of Zürich
- 2019: Main organizer of a series of events for the **promotion of Diversity in Science**, U. of Zürich; main speaker and moderator in one meeting ([https://padlet.com/eli\\_bortolas/39e0898pr0uu](https://padlet.com/eli_bortolas/39e0898pr0uu)).
- 2013 – 2015: Student representative of the Astrophysics and Space Physics master students, U. of Milano-Bicocca.

## IT experience

- *AI tools*: Claude Code, ChatGPT; effective use of LLMs for coding, debugging, and research workflows
- *Programming Languages*: **python** (advanced), **C** (advanced), C++ (good), bash (good)
- *Semi-analytical models for galaxy evolution*: L-galaxies-BH
- *Astrophysics related numerical integrators*: HiGPUs, phiGRAPE, phiGPU, gasoline2, Phaseflow, AGAMA, PeTar
- *Science-related software/packages*: ROOT, IRAF, SPLASH; python: Numpy, Scipy, pynbody
- *Job Schedulers*: PBS, Slurm
- Extensive experience in GPU computing and parallel computing (CUDA, OpenCL; openMP, MPI)
- Experience in code optimization and high performance computing

## Memberships

- Member of the consortium for the development of the **Lunar Gravitational Wave Antenna**, a detector based on the idea of Weber to monitor low-frequency vibrations of the Moon excited by gravitational waves
- Member of the **Einstein Telescope Observational Science Board** – *Division 3: Population studies*
- Member of the Istituto Nazionale di Fisica Nucleare (INFN, National Institute of Nuclear Physics)
- Member of the Italian **TEONGRAV** collaboration – *Teoria delle onde gravitazionali* (theory of gravitational waves)
- Member of the **European Pulsar Timing Array (EPTA)** for the detection of low-frequency gravitational waves
- Full member of the **LISA Consortium**
- Member of the **LISA Early Career Scientist** Group (LECS) and the **Gravitational Waves Early Career Scientist** Group (GWECS)
- Contributor to the White Book for the scientific cases of **MORFEO** (previously known as MAORY), the multi-conjugate adaptive optics module of the European Extremely Large Telescope.

## Participation in advanced courses and schools (highlights)

- *International HPC Summer School 2018*, **highly competitive admission** (80 participants selected among ~600 applicants); July 8–13, 2018 in Ostrava (Czech Republic)
- 11<sup>th</sup> TRR33 Winter School in Cosmology, *Theory for Observers & Observations for Theorists*, December 10–16, 2017, Passo del Tonale (Italy); poster presentation *Supernova Kicks and Dynamics of Compact Remnants in the Galactic Centre*, **first prize for the best poster of the school**

- CINECA school 25<sup>th</sup> Summer School on Parallel Computing, June 6–17, 2016 in Casalecchio di Reno, Bologna (Italy)

### Outreach activities

- **Public talk for the 25<sup>th</sup> Anniversary of the University of Milano-Bicocca** (*Amazing Physics/Fisica stupefacente*), April 20, 2023; talk title: *Fenomeni transienti ad alta energia per censire i buchi neri di massa intermedia* (*High energy transients as sensors of intermediate mass black holes*);
- Support for the series of events *Studiare fisica: conoscere, sperimentare, innovare*, an outreach event to engage high school students with the research in Astrophysics, University of Milano-Bicocca, 2023;
- Outreach seminar for the Italian *AstronomiAmo* Association, 2020;
- Public outreach seminar at the Auditorium of Varano Borghi (Italy), 2020;
- Public outreach talk for the Livio Gratton Prize in Frascati (Italy), 2019;
- Outreach seminar at the Padova Planetarium, 2018;
- Member of the committee for the organization of *Pint of Science* 2018 in Padova (Italy);
- *Notte dei Ricercatori* (The night of researchers) 2017, public outreach events in Padova (Italy);
- Support for the monthly public star gazing events held at the University of Surrey telescope, 2016 – 2017.

### Press coverage

- INAF press release for the award of the Gratton Prize 2019: [www.media.inaf.it/2019/07/18/il-premio-gratton-a-una-tesi-sui-buchi-neri/](http://www.media.inaf.it/2019/07/18/il-premio-gratton-a-una-tesi-sui-buchi-neri/)
- INAF press release for the award of the Magini Prize 2017: [www.media.inaf.it/2017/05/29/premio-magini-arcetri/](http://www.media.inaf.it/2017/05/29/premio-magini-arcetri/)

### Language knowledge

Italian (native), English (fluent), French (basic), Spanish (basic).

## PUBLICATIONS

### Publications in peer-reviewed journals

**47 published papers** in high-impact, peer-reviewed astronomy journals; 2 submitted papers.

total citations: 5388

h-index: 27

source: SAO/NASA Astrophysics Data System (ADS), October 2026

Among the published papers:

- 11 publications as leading (first) author, among which one as single author
- 14 publications led by students I mentored
- 41 publications without PhD advisor

The symbol • identifies students I advised towards the development of the project that led to the publication.

47. Saracco, P., et al. (including **Bortolas, E.**), *SHARP – A spectrograph proposal to fully exploit ELT capabilities and look beyond JWST*, 2026, NewA 129, 102619

[<https://ui.adsabs.harvard.edu/abs/2026NewA..12902619S>]

46. **Bortolas, E.**, Portaluri, E., Dalla Bontà, E., Arcidiacono, C., Severgnini, P., Gualandris, A., *Unveiling the properties of galaxy cores excavated by supermassive black hole binaries with SHARP*, 2026, NewA 128, 102593

[<https://ui.adsabs.harvard.edu/abs/2026NewA..12802593B>]

45. Dalla Bontà, E., Portaluri, E., Arcidiacono, C., **Bortolas, E.**, et al., *Supermassive black holes and nuclear star clusters in galaxies: Constraining formation and co-evolution with SHARP and MICADO*, 2026, NewA 128, 102613

[<https://ui.adsabs.harvard.edu/abs/2026NewA..12802613D>]

44. Severgnini, P., et al. (including **Bortolas, E.**), *A new era for dual AGN science with SHARP*, 2026, NewA 128, 102610

[<https://ui.adsabs.harvard.edu/abs/2026NewA..12802610S>]

43. Einstein Telescope collaboration (including **Bortolas, E.**), *The Science of the Einstein Telescope* (this is the blue book of the Einstein telescope), 2026, JCAP 03, 081

[<https://ui.adsabs.harvard.edu/abs/2026JCAP...03..081A>]

42. Fastidio, F. <sup>•</sup>, **Bortolas, E.**, Gualandris, A., Sesana, A., Read, J. I., Dehnen, W., *Realistic consecutive galaxy mergers form eccentric pulsar timing array sources*, 2025, A&A 703, A86  
[https://ui.adsabs.harvard.edu/abs/2025A%26A...703A..86F]
41. Porayko, N., et al. (EPTA collaboration including **Bortolas, E.**), *Searches for signatures of ultra-light axion dark matter in polarimetry data of the European Pulsar Timing Array*, 2025, PhRvD 111, 062005  
[https://ui.adsabs.harvard.edu/abs/2025PhRvD.111f2005P]
40. LGWA collaboration (including **E. Bortolas**), *The Lunar Gravitational-wave Antenna: Mission Studies and Science Case*, 2025, JCAP 1, 108. **I coordinated the drafting of sec. 3.2.2**  
[https://ui.adsabs.harvard.edu/abs/2024arXiv240409181A/]
39. Langen, V. <sup>•</sup>, Tamanini, N., Marsat, S., **Bortolas, E.**, *Hierarchical Bayesian inference on an analytical toy model of the LISA massive black hole binary population*, 2025, MNRAS 536, 3366  
[https://ui.adsabs.harvard.edu/abs/2025MNRAS.536.3366L]
38. EPTA collaboration (including **E. Bortolas**), *The second data release from the European Pulsar Timing Array V. Search for continuous gravitational wave signals*, 2024, A&A 690, A118  
[https://ui.adsabs.harvard.edu/abs/2024A%26A...690A.118E]
37. Varisco, L. <sup>•</sup>, Dotti, M., Bonetti, M., **Bortolas, E.**, Lupi, A., *An effective model for the tidal disruption of satellites undergoing minor mergers with axisymmetric primaries*, 2024, A&A 689, A279  
[https://ui.adsabs.harvard.edu/abs/2024A%26A...689A.279V]
36. Polkas, M. <sup>•</sup>, Bonoli, S., **Bortolas, E.**, Izquierdo-Villalba, D., Sesana, A., Broggi, L., Hoyer, N., Spinoso, D., *Demographics of Tidal Disruption Events with L-Galaxies: I. Volumetric TDE rates and the abundance of Nuclear Star Clusters*, 2024 A&A 689, 204 (Corrigendum: 2026, A&A 712, C7)  
[https://ui.adsabs.harvard.edu/abs/2024A%26A...689A.204P/abstract]
35. Broggi, L. <sup>•</sup>, Stone, N. C., Ryu, T., **Bortolas, E.**, Dotti, M., Bonetti, M., Sesana, A., *Repeating partial disruptions and two-body relaxation*, 2024, OJA , 7, 48 [https://ui.adsabs.harvard.edu/abs/2024OJAp....7E..48B/]
34. Fastidio, F. <sup>•</sup>, Gualandris, A., Sesana, A., **Bortolas, E.**, Dehnen, W., *Eccentricity evolution of PTA sources from cosmological initial conditions*, 2024, MNRAS, 532, 295.  
[https://ui.adsabs.harvard.edu/abs/2024MNRAS.532..295F]
33. EPTA collaboration (including **E. Bortolas**), *Comparing Recent Pulsar Timing Array Results on the Nanohertz Stochastic Gravitational-wave Background*, 2024, ApJ 966, 105.  
[https://ui.adsabs.harvard.edu/abs/2024ApJ...966..105A]
32. EPTA collaboration (including **E. Bortolas**), *The second data release from the European Pulsar Timing Array. IV. Implications for massive black holes, dark matter, and the early Universe*, 2024, A&A 685, 94.  
[https://ui.adsabs.harvard.edu/abs/2024A%26A...685A..94E/]
31. Tang, V. L., Madau, P., **Bortolas, E.**, Peng, E. W., Feng, Y., Guhathakurta, P., *Searching for Intermediate-mass Black Holes in Globular Clusters through Tidal Disruption Events*, 2024, ApJ 963, 146  
[https://ui.adsabs.harvard.edu/abs/2024ApJ...963..146T]
30. EPTA collaboration (including **E. Bortolas**), *Practical approaches to analyzing PTA data: Cosmic strings with six pulsars*, 2023, PhRvD 108, 12, 123527.  
[https://ui.adsabs.harvard.edu/abs/2023PhRvD.108l3527Q]
29. LISA Astrophysics Working Group, *Astrophysics with the Laser Interferometer Space Antenna (LISA Astrophysics White Paper)*, 2023, LRR 26, 2. **I coordinated the drafting of chapter 2 (Massive Black Holes)** and I contributed as an author to chapter 2 and 3.  
[https://ui.adsabs.harvard.edu/abs/2023LRR....26....2A]
28. EPTA collaboration (including **E. Bortolas**), *Second Data Release from the European Pulsar Timing Array: Challenging the Ultralight Dark Matter Paradigm*, 2023, PhRvL 131, 17, 171001.  
[https://ui.adsabs.harvard.edu/abs/2023PhRvL.131q1001S]
27. EPTA collaboration (including **E. Bortolas**), *The second data release from the European Pulsar Timing Array. I. The dataset and timing analysis*, 2023, A&A 678, 48.  
[https://ui.adsabs.harvard.edu/abs/2023A%26A...678A..48E/]
26. EPTA collaboration (including **E. Bortolas**), *The second data release from the European Pulsar Timing Array. II. Customised pulsar noise models for spatially correlated gravitational waves*, 2023, A&A 678, 49.  
[https://ui.adsabs.harvard.edu/abs/2023A%26A...678A..49E]

25. EPTA collaboration (including **E. Bortolas**), *The second data release from the European Pulsar Timing Array. III. Search for gravitational wave signals*, 2023, A&A 678, 50.  
[https://ui.adsabs.harvard.edu/abs/2023A%26A...678A..50E]
24. **Bortolas, E.**, Ryu, T., Broggi, L. <sup>•</sup>, Sesana, A., *Partial stellar tidal disruption events and their rates*, 2023, MNRAS 524, 3026  
[https://ui.adsabs.harvard.edu/abs/2023MNRAS.524.3026B]
23. Franchini, A., Bonetti, M., Lupi, A., Miniutti, G., **Bortolas, E.**, et al., *Quasi periodic eruptions from impacts between the secondary and a rigidly precessing accretion disc in an EMRI system*, 2023, A&A 675, 100  
[https://ui.adsabs.harvard.edu/abs/2023A%26A...675A.100F]
22. Dotti, M., Bonetti, M., Rigamonti, F., **Bortolas, E.**, Fossati, M., et al., *Optical follow-up of the tick-tock massive black hole binary candidate*, 2023, MNRAS 518, 4172  
[https://ui.adsabs.harvard.edu/abs/2023MNRAS.518.4172D]
21. Broggi, L. <sup>•</sup>, **Bortolas, E.**, Bonetti, M., Sesana, A., Dotti, M., *Extreme mass ratio inspirals and tidal disruption events in nuclear clusters. I. Time dependent rates*, 2022, MNRAS 514, 3270  
[https://ui.adsabs.harvard.edu/abs/2022arXiv220506277B]
20. **Bortolas, E.**, Bonetti, M., Dotti, M., Lupi, A., Capelo, P.R., Mayer, L. and Sesana, A., *The role of bars on the dynamical-friction driven inspiral of massive perturbers*, 2022, MNRAS, 512, 3365  
[https://ui.adsabs.harvard.edu/abs/2022MNRAS.512.3365B]
19. **Bortolas, E.**, *Tidal disruption events in poststarburst galaxies: the importance of a complete stellar mass function*, 2022, MNRAS 511, 2885 (**single-authored**)  
[https://ui.adsabs.harvard.edu/abs/2022MNRAS.511.2885B/]
18. Gualandris, A., Khan, F., **Bortolas, E.**, Bonetti, M., Sesana, A., Berczik, P. and Holley-Bockelmann, K., *Eccentricity evolution of massive black hole binaries from formation to coalescence*, 2022, MNRAS 511, 4753  
[https://ui.adsabs.harvard.edu/abs/2022MNRAS.511.4753G]
17. Vázquez-Aceves, V. <sup>•</sup>, Zwick, L. <sup>•</sup>, **Bortolas, E.**, Capelo, P.R., Amaro-Seoane, P., Mayer, L., and Xian, C., *Revised event rates for extreme and extremely large mass-ratio inspirals*, 2022, MNRAS 510, 2379  
[https://ui.adsabs.harvard.edu/abs/2022MNRAS.510.2379V]
16. Varisco, L. <sup>•</sup>, **Bortolas, E.**, Dotti, M. and Sesana, A., *Stellar hardening of massive black hole binaries: the impact of the host rotation*, 2021, MNRAS 508, 1533  
[https://ui.adsabs.harvard.edu/abs/2021MNRAS.508.1533V]
15. Zwick, L. <sup>•</sup>, Capelo, P.R., **Bortolas, E.**, Vázquez-Aceves, V., Mayer, L. and Amaro-Seoane, P., *Improved gravitational radiation time-scales II: spin-orbit contributions and environmental perturbations*, 2021, MNRAS, 506, 1007  
[https://ui.adsabs.harvard.edu/abs/2021MNRAS.506.1007Z]
14. **Bortolas, E.**, Franchini, A., Bonetti, M., Sesana, A., *The Competing Effect of Gas and Stars in the Evolution of Massive Black Hole Binaries*, 2021, **ApJ Letter** 918, 15  
[https://ui.adsabs.harvard.edu/abs/2021ApJ...918L..15B]
13. Bonetti, M., **Bortolas, E.**, Lupi, A., Dotti, M., *Dynamical evolution of massive perturbers in realistic multi-component galaxy models I: implementation and validation*, 2021, MNRAS, 502, 3554  
[https://ui.adsabs.harvard.edu/abs/2021MNRAS.502.3554B]
12. Pestoni, B. <sup>•</sup>, **Bortolas, E.**, Capelo, P.R., Mayer, L., *Generation of gravitational waves and tidal disruptions in clumpy galaxies*, 2021, MNRAS, 500, 4628  
[https://ui.adsabs.harvard.edu/abs/2021MNRAS.500.4628P/]
11. **Bortolas, E.**, Capelo, P.R., Zana, T., Mayer, L., Bonetti, M., Dotti, M., Davies, M.B., Madau, P., *Global torques and stochasticity as the drivers of massive black hole pairing in the young Universe*, 2020, MNRAS, 498, 3601  
[https://ui.adsabs.harvard.edu/abs/2020MNRAS.498.3601B]
10. Souza Lima, R. <sup>•</sup>, Mayer, L., Capelo, P.R., **Bortolas, E.**, Quinn, T.R., *The erratic path to coalescence of LISA massive black hole binaries in sub-pc resolution simulations of smooth circumnuclear gas disks*, 2020,

ApJ, 899, 126

[<https://ui.adsabs.harvard.edu/abs/2020ApJ...899..126S>]

9. Zwick, L. <sup>•</sup>, Capelo, P. R., **Bortolas, E.**, Mayer, L., Amaro-Seoane, P., *Improved gravitational radiation time-scales: significance for LISA and LIGO-Virgo sources*, 2020, MNRAS, 495, 2321  
[<https://ui.adsabs.harvard.edu/abs/2020MNRAS.495.2321Z>]
8. Bonetti, M., **Bortolas, E.**, Lupi, A., Dotti, M., Raimundo, S.I., *Dynamical friction-driven orbital circularisation in rotating discs: a semi-analytical description*, 2020, MNRAS, 494, 3053  
[<https://ui.adsabs.harvard.edu/abs/2020MNRAS.494.3053B>]
7. Stegmann, J. <sup>•</sup>, Capelo, P. R., **Bortolas, E.**, Mayer, L., *Improved constraints from ultra-faint dwarf galaxies on primordial black holes as dark matter*, 2020, MNRAS, 492, 5247  
[<https://ui.adsabs.harvard.edu/abs/2020MNRAS.492.5247S>]
6. **Bortolas, E.** & Mapelli, M. *Can Supernova Kicks trigger EMRIs in the Galactic Centre?*, 2019, MNRAS, 485, 2125  
[<https://ui.adsabs.harvard.edu/abs/2019MNRAS.485.2125B>]
5. **Bortolas, E.**, Gualandris, A., Dotti, M., Read, J. I., *The influence of Massive Black Hole Binaries on the Morphology of Merger Remnants*, 2018, MNRAS 477, 2310  
[<https://ui.adsabs.harvard.edu/abs/2018MNRAS.477.2310B>]
4. **Bortolas, E.**, Mapelli, M., Spera, M., *Star cluster disruption by a massive black hole binary*, 2018, MNRAS 474, 1054  
[<https://ui.adsabs.harvard.edu/abs/2018MNRAS.474.1054B>]
3. **Bortolas, E.**, Mapelli, M., Spera, M., *Supernova Kicks and Dynamics of Compact Remnants in the Galactic Centre*, 2017, MNRAS 469, 1510  
[<https://ui.adsabs.harvard.edu/abs/2017MNRAS.469.1510B>]
2. Gualandris, A., Read, J. I., Dehnen, W., **Bortolas, E.**, *Collisionless loss-cone refilling: there is no final parsec problem*, 2017, MNRAS, 464, 2301  
[<https://ui.adsabs.harvard.edu/abs/2017MNRAS.464.2301G>]
1. **Bortolas, E.**, Gualandris, A., Dotti, M., Spera, M., Mapelli, M., *Brownian motion of massive black hole binaries and the final parsec problem*, 2016, MNRAS 461, 1023  
[<https://ui.adsabs.harvard.edu/abs/2016MNRAS.461.1023B>]

### Papers submitted for publication in peer-reviewed journals

- Capelo, P. R., Moreno Martinez, C., Koren, N., Zana, T., **Bortolas, E.**, Bonetti, M., Fluri, J., Hofmann, T., Mayer, L., *Emulating the complex galactic-scale orbital dynamics of LISA massive black hole pairs with normalizing flows*, 2026, submitted  
[<https://ui.adsabs.harvard.edu/abs/2026arXiv260726125C>]
- Izquierdo-Villalba, D., et al. (including **Bortolas, E.**), *The LISA Astrophysics MBHcatalogues Project: A comparison of predictions of simulated massive black hole binaries*, 2026, submitted  
[<https://ui.adsabs.harvard.edu/abs/2026arXiv260500092I>]

### Conference proceedings

3. Ballone, A., Mapelli, M., **Bortolas, E.**, Trani, A.A., *The inner parsecs of our Galaxy: star formation and its environment*, *Astrophysics with the INAF-CINECA MoU 2017-2020*, F. Pitari, G. Mulas, G. Murante, P. Rossi, M.S. Cremonesi, M. Guarrasi, C. Padrin and U. Becciani eds, Mem. SAI, Vol 90  
[<https://ui.adsabs.harvard.edu/abs/2020MmSAI..91..201B>]
2. **Bortolas, E.**, Mapelli, M., Spera, M., *Star Cluster Disruption by a Supermassive Black Hole Binary*, *Proceedings of Science* (proceedings of the GRAVitational-waves Science & technology Symposium - GRASS2018, 1-2 March 2018, Palazzo Moroni, Padova, Italy) [<https://ui.adsabs.harvard.edu/abs/2018gwss.confE..30B>]
1. **Bortolas, E.**, Mapelli, M., Spera, M., *Dynamics of supernova remnants in the Galactic Centre*, *Cosmic-Lab: Star Clusters as Cosmic Laboratories for Astrophysics, Dynamics and Fundamental Physics*, F.R. Ferraro and B. Lanzoni eds, Mem. SAI, Vol 87 (proceedings of the Cosmic-Lab conference, 18-22 April 2016, Bologna, Italy) [<https://ui.adsabs.harvard.edu/abs/2016MmSAI..87..679B>]

## Theses

- November 2018: **Bortolas, E.**, *Dynamics of Single and Binary Black Holes in Galactic Nuclei*, PhD Thesis in Astronomy [<http://paduaresearch.cab.unipd.it/11825>]; **the thesis was awarded the prestigious Gratton Prize**
- September 2015: **Bortolas, E.**, *Dynamics of Massive Black Hole Binaries in Galactic Merger Remnants*, Master Thesis in Astrophysics and Space Physics [<https://drive.google.com/open?id=0Bx7KfRPRjKbZTRkc19yeXdLc1E>]; **the thesis was awarded the prestigious Magini Prize**